AWS redshift

AWS redshift is a distributed relational cloud data warehouse. Distributed means that Redshift can distribute the large data sets that you have to multiple machines and run queries on those data sets in parallel, utilizing CPU and memory of multiple computers. Relational means that its core Redshift is based on a relational technology. In conclusion, cloud data warehouse means that a lot of management operations on the warehouse performed by AWS and are hidden from the end users. A redshift cluster consist of different clusters. A leader node is responsible for accepting all connections from clients, parsing queries and distributing queries to the corresponding nodes. A leader is also responsible for deciding which node will get the new data as it is added into the data warehouse.

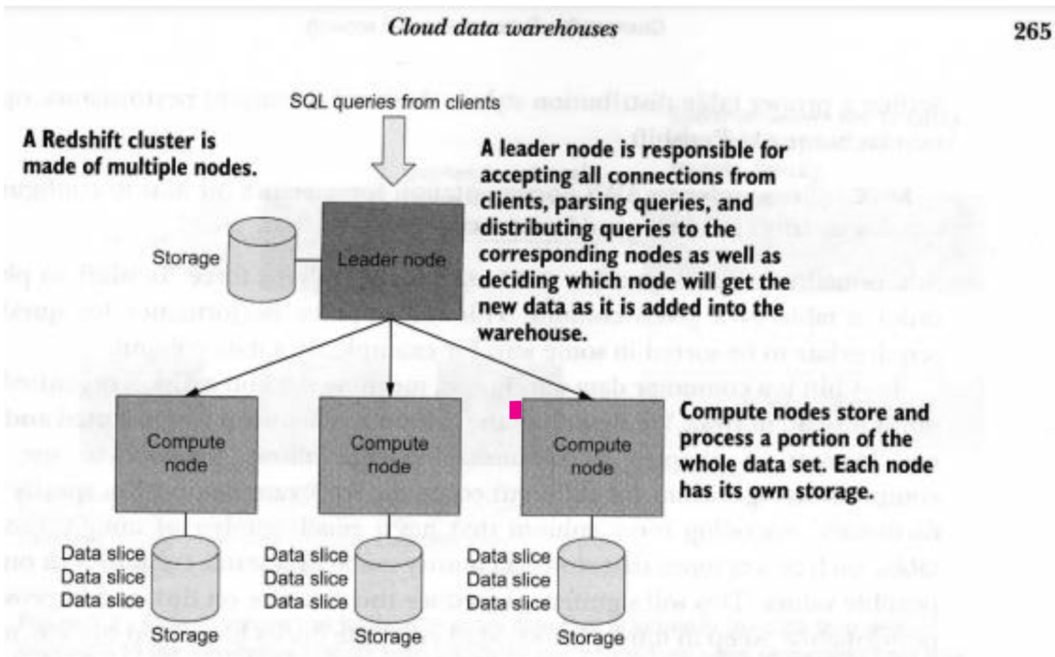

Figure 9.2 An AWS Redshift cluster consists of a leader node with multiple compute nodes that are used to distribute data and workload among themselves. Each node has its own storage attached to it.

In redshift, you can also create the external table instead of creating a new node. For example, you have contained new demographic information for your customer in order to make better decision in your business users. This new data set require a new node and it will consume data storage and compute resources. On the other hand, you can also use the external tables and use spectrum to run external queries outside Redshift. Thus end users can discard the data set if they don't find it helpful. We recommend you group external tables into a database and periodically clean up tables. In our data platform architecture, spectrum is a definitely an optional feature, because you can achieve the same result by running Spark jobs or Spark SQL queries.

Azure Synapse

Azure synapse is a distributed cloud data warehouse offering from Microsoft. It is based on parallel data warehouse product and is based on a relational technology.

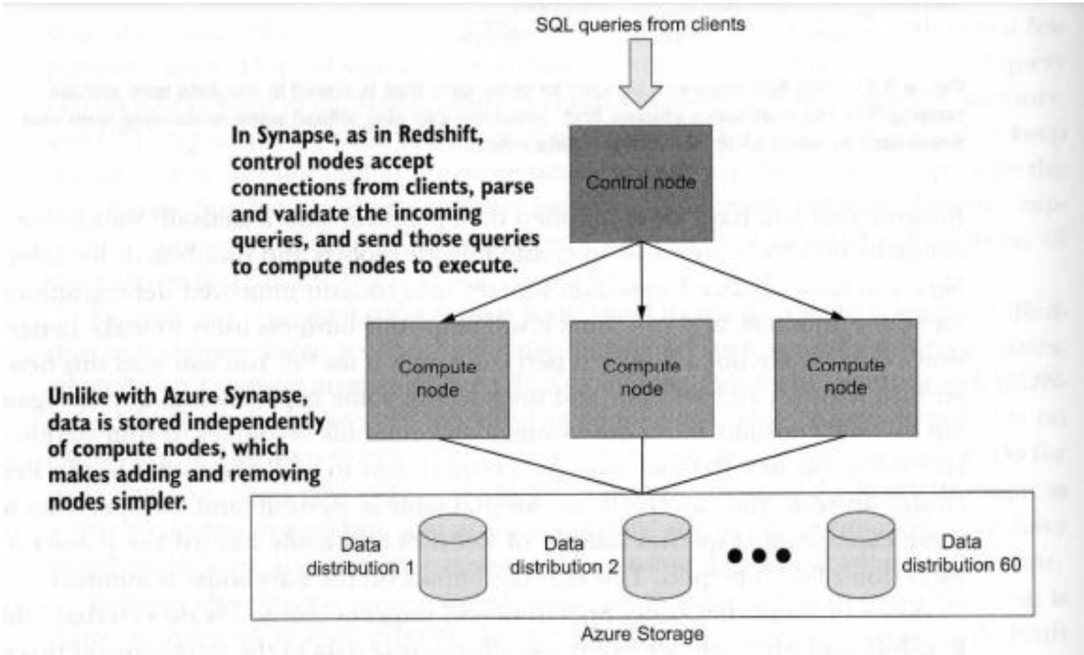

Figure 9.4 An Azure Synapse cluster consists of a control node and compute nodes, with data storage being completely separate from the compute layer.

Synapse clusters consists of control node and compute node. The control node accepts connections from clients, parses and validates the incoming queries, and send those queries to compute nodes to execute. Synapse separates the storage layer from the compute layer. Synapse splits all your data into 60 data distributions, and each data distribution is attached to a particular compute node, but data is not stored on compute nodes themselves. Scaling a synapse cluster is an offline operation, meaning cluster will need to complete or cancel any running queries and will not accept any new queries while the scaling operation is underway

One major difference between redshift and synapse is that synapse is separates the storage layer into compute layer. Synapse splits all your data into 60 data distributions, and each data distribution is attached to a particular compute node, but data is not stored on the compute node themselves. This design makes scaling the cluster a simple and fast operation. Synapse operation is an offline operation, meaning the cluster will need to complete or cancel any running queries and will not accept any new queries while the scaling operation is underway. This makes synapse not truly elastically scalable. Another property of design of this architecture is you can pause the data anytime and resume them without having move the data anywhere.

Azure synapse also supports apache spark pools.

Google BigQuery:

BigQuery is closer to fully managed service than synapse and redshift. The cluster capacity is provisioned by Google cloud on the fly. It will be different from query to query. One of the side effects of Bigquery for non relational database is that it is not seamlessly compatible with visualization tools

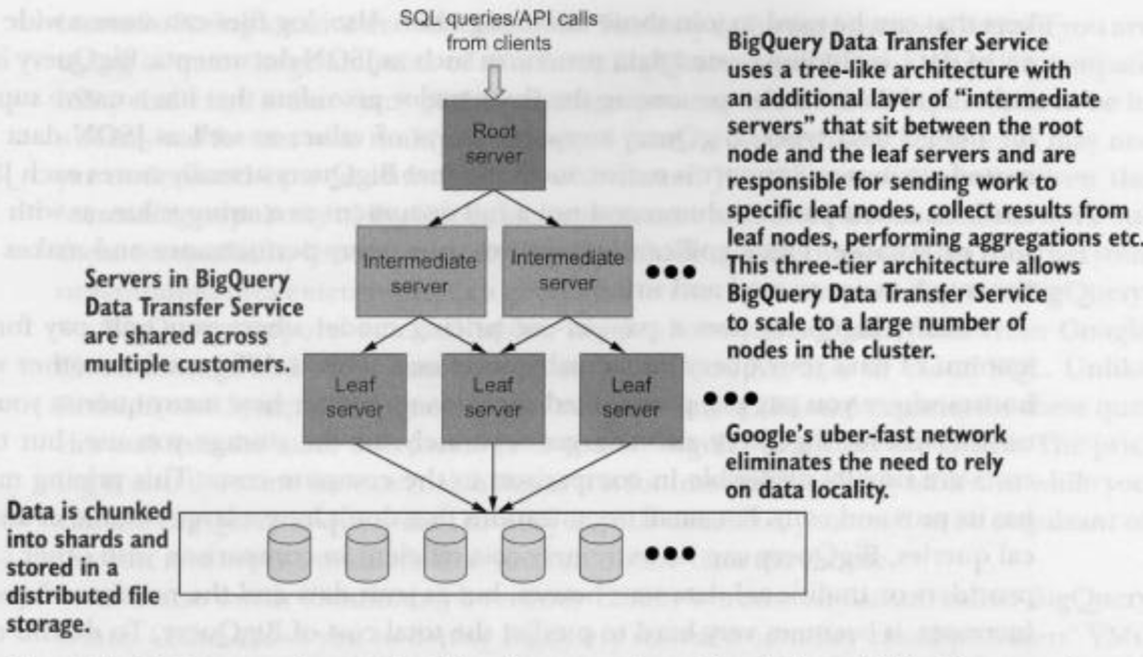

Figure 9.5 BigQuery uses large pools of shared resources to provide elastic scalability. Storage and compute are separate, and the fast internal Google network allows not to worry about data and compute nodes being "close."

Differences:

- Azure synapse uses a columnar data store and applies compression to each column separately. Unlike with Redshift, you can't specify a compression algorithm for each column, and you have to rely on synapse to make the best choice for you.
- Bigquery is closer to fully managed service than Synapse and Redshift. Having a three tier architecture instead of a two tier architecture like we saw in Redshift and Synapse allows BigQuery to scale to a much larger number of nodes in the cluster.
- One major difference between redshift and synapse is that synapse is separates the storage layer into compute layer. Synapse splits all your data into 60 data distributions, and each data distribution is attached to a particular compute node, but data is not stored on the compute node themselves.

Table 9.1 Comparing key data warehouse features

	AWS Redshift	Azure Synapse	Google BigQuery
Based on a relational technology?	Yes	Yes	No
Support for nested data structures?	Limited (via JSON parsing functions)	Limited (via JSON parsing functions)	Native support
How does it scale up/down?	Manually	Manually	Automatically
Pay per use or pay per provisioned capacity?	Pay per provisioned capacity	Pay per provisioned capacity	Pay per use (with option to pay per provisioned capacity)

Takeaways:

- The large organization uses different cloud data warehouse according to their requirement.
- AWS redshift is a distributed relational cloud data warehouse. Distributed means that Redshift can distribute the large data sets that you have to multiple machines and run queries on those data sets in parallel, utilizing CPU and memory of multiple computers. It consists of nodes. In redshift, you can also create the external table instead of creating a new node. The external tables are segregated so it is recommended that should use group external table and remove unnecessary tables when it has been utilized.
- Azure synapse is a distributed cloud data warehouse offering from Microsoft. It is based on parallel data warehouse product and is based on a relational technology.
- Synapse clusters consists of control node and compute node.
- **BigQuery** is closer to fully managed service than synapse and redshift. The cluster capacity is provisioned by Google cloud on the fly. It will be different from query to query.